How-To Library

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Data Vault mapping task template configuration tutorial

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Abstract

Data Integration provides parameterized mapping task templates to help you organize your data into Data Vault hubs, links, and satellites. This article explains how to configure each template to generate unique business keys and dynamically load new and changed data into hub, link, and satellite tables.

Supported Versions

- Data Integration October 2025

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Data Vault mapping task template overview

You can use a mapping task template to organize your data into Data Vault hub, link, and satellite tables instead of creating a mapping from scratch. Each task template is based on a parameterized mapping.

In this tutorial we use the following templates:

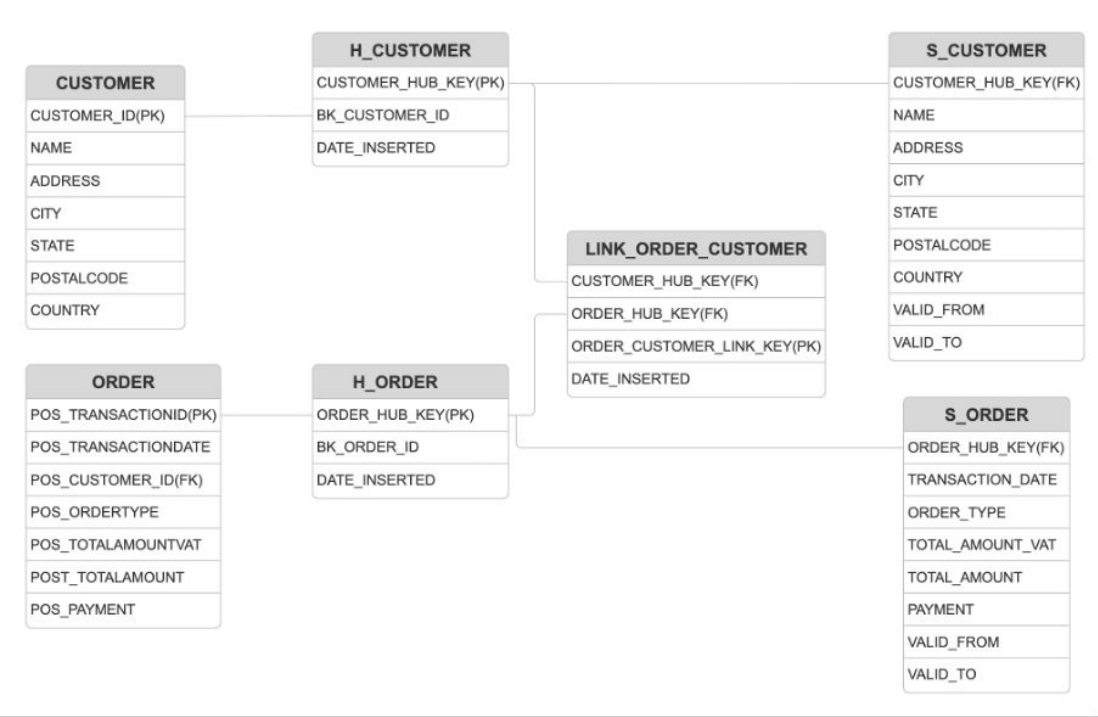
- Data Vault – Hub. Generates unique business keys and writes them to a hub table along with the date the records are loaded.
- Data Vault – Link. Creates relationships between hub or other link tables based on business keys and stores the connection in a link table.
- Data Vault – Satellite. Generates a satellite table to store descriptive data such as context, history, details, for either a hub or a link.

About the tutorial example

This tutorial describes the steps to configure parameterized Data Vault task templates available in Data Integration. For this tutorial we have raw customer data that we want to link to order data while tracking changes to each record. To do this, we use the Data Vault mapping task templates.

You have a landing table in a Snowflake Data Cloud database that contains raw customer data. You need to generate business keys in the landing table and load them to a hub table, H_CUSTOMER. You then need to link the H_CUSTOMER hub with the hub table H_ORDERS, and load the relationship to the link table LINK_ORDER_CUSTOMER. Finally, you want to load customer metadata to the satellite table S_CUSTOMER and track historical changes to the data.

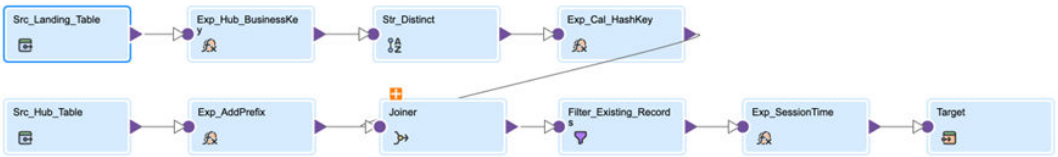
The following image shows the table structure that we create in this tutorial:



Step 1. Load business keys to a hub table

In the source landing table, the CUSTOMER_ID field is the primary key. We want to hash this value and write both keys to the H_CUSTOMER hub table along with the date the records are inserted. To do this, we use the Data Vault – Hub task template.

The following image shows the mapping this template is based on:



When the task runs, it hashes the incoming primary key and uses the hash attribute as the primary key in the hub table.

1. On the **Templates** page, click the **Data Vault – Hub** tile and then click **Create a mapping task**.
2. On the **General** tab, select the runtime environment to run the task and then click **Next**.
3. On the **Sources** tab, select the landing table connection and object for the Src_Landing_Table source.



4. Select the connection and hub table for the Src_Hub_Table source.

   The template is preconfigured to check if records are already present in the target hub table.



5. Click **Next**.
6. On the **Target** tab, select the target connection and the target hub table.



7. Select the **Insert** operation.
8. On the **Input parameters** page, resolve the following parameters:
   a. Hub Key from Landing Table. Defines the primary key in the landing table. You define the CUSTOMER_ID field as the primary key using the following expression:

      ```
      TO_CHAR(TRUNC(CUSTOMER_ID))
      ```

   b. Filter Existing Records. Filters out records that already exist in the hub table so only new records get inserted. You enter the following expression:

      ```
      IIF(ISNULL(BK_CUSTOMER_ID),TRUE, FALSE)
      ```

   c. Join Condition. Joins the landing table and the hub table by the CUSTOMER_ID field. You specify the join condition using the following condition:

      ```
      Landing_Hub_Key = BK_CUSTOMER_ID
      ```

   d. Hub Field Mapping. The following table shows the field mapping configuration to write data to the target:

Target fields	Incoming fields
CUSTOMER_HUB_KEY	Landing_Hash_Key
EX_CUSTOMER_ID	Landing_Hub_Key
DATE_INSERTED	DATE_INSERTED

9. Click **Save**.

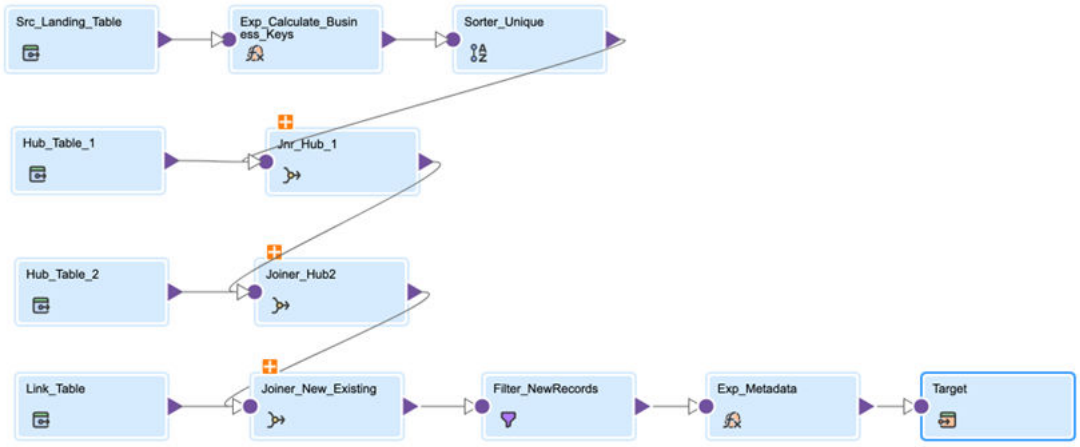
When the task runs, it loads the new business keys into the hub table along with the hash key and the date when the record is loaded as shown in the following image:

	CUSTOMER_HUB_KEY	BK_CUSTOMER_ID	DATE_INSERTED
1	C4CA4238A0B923820DCC509A	1	2025-07-29 11:49:05.000
2	C81E728D9D4C2F636F067F89C	2	2025-07-29 11:49:05.000
3	ECCBC87E4B5CE2FE28308FD9I	3	2025-07-29 11:49:05.000
4	A87FF679A2F3E71D9181A67B7E	4	2025-07-29 11:49:05.000
5	E4DA3B7FBBCE2345D7772B067	5	2025-07-29 11:49:05.000
6	1679091C5A880FAF6FB5E6087I	6	2025-07-29 11:49:05.000

Step 2. Configure customer-order relationships in a link table

We need to define customer and order relationships and write the business keys to a link table. To do this we use the Data Vault – Link mapping task template.

The following image shows the mapping this template is based on:



When the task runs, it calculates customer and order hash keys, and a primary hash key for the link table. It then writes the keys and the date inserted to the target.

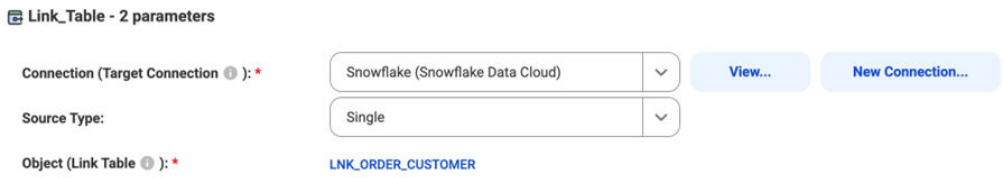
1. On the **Templates** page, click the **Data Vault – Link** tile and then click **Create a mapping task**.
2. On the **General** tab, select the runtime environment to run the task and then click **Next**.
3. On the **Sources** tab, configure the following sources:
   a. Hub_Table_1 source. Select the Snowflake connection and H_ORDER hub table.



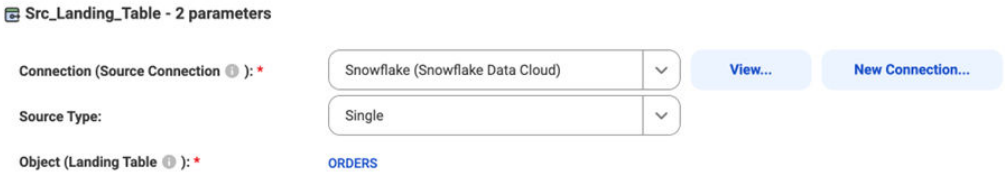
   b. Hub_Table_2 source. Select the Snowflake connection and H_CUSTOMER hub table.



   c. Link_Table. The template is preconfigured to check if records are already present in the target link table. Select the target Snowflake connection and LNK_ORDER_CUSTOMER link table.



d. Src_Landing_Table. Select the Snowflake connection and the ORDER landing table.



4. On the **Targets** tab, select the Snowflake target connection and link table, LNK_ORDER_CUSTOMER.



5. Select the **Insert** operation.
6. On the **Input Parameters** tab, resolve the following parameters:
   a. Hub Key Expression 1. Defines the hash key in the ORDER landing table and outputs the value to the Hub_Key_Calculated1 field. You assign the hash key to the POS_TRANSACTIONID field using the following expression:

      ```
      MD5(TO_CHAR(TRUNC(POS_TRANSACTIONID)))
      ```

   b. Hub Key Expression 2. Defines the customer hash key and outputs the value to the Hub_Key_Calculated2 field. You assign the hash key to the POS_CUSTOMER_ID field using the following expression:

      ```
      MD5(LTRIM(RTRIM(POS_CUSTOMER_ID)))
      ```

   c. Hub 1 – Join Condition: Defines the join condition between the H_ORDER hub table and the hub order key we calculated in step 6a. You enter the following expression:

      ```
      Hub_Key_Calculated1 = Hub1_ORDER_HUB_KEY
      ```

   d. Hub 2 – Join Condition. Defines the join condition between the H_CUSTOMER hub table and the hub customer key we calculated in step 6b. You enter the following expression:

      ```
      Hub_Key_Calculated2 = Hub2_CUSTOMER_HUB_KEY
      ```

   e. Join Condition New Record. Joins new records with existing data. You enter the following expression:

      ```
      Hub_Key_Calculated1 = Link_ORDER_HUB_KEY AND Hub_Key_Calculated2 = Link_CUSTOMER_HUB_KEY
      ```

   f. Link Key: Generates the hash key for the link table. Concatenate the customer and order keys with the following expression:

      ```
      MD5(Concat(Hub1_BK_ORDER_ID,Hub2_BK_CUSTOMER_ID))
      ```

   g. Target Field Mapping. Defines the field mapping between incoming fields and target fields. The following table shows how to configure the field mapping:

Target field	Incoming field
ORDER_HUB_KEY	Hub_Key_Calculated1
CUSTER_HUB_KEY	Hub_Key_Calculated2

Target field	Incoming field
DATE_INSERTED	DATE_INSERTED
ORDER_CUSTOMER_LINK_KEY	LINK_KEY

7. Click **Save**.

When the task runs, it writes new customer-order relationships in the link table as shown in the following image:

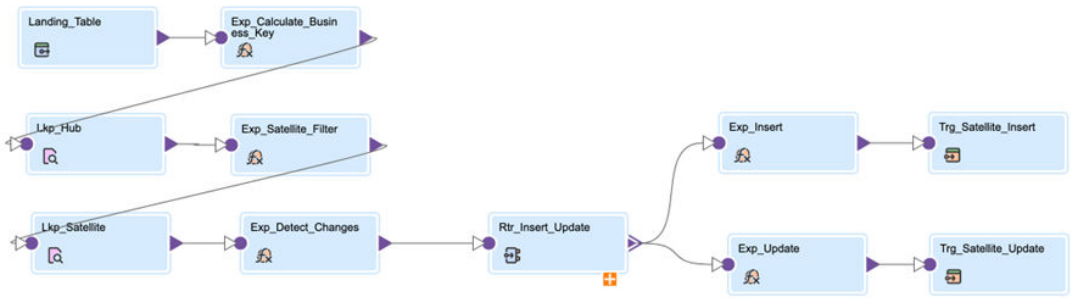
	ORDER_HUB_KEY	CUSTOMER_HUB_KEY	DATE_INSERTED	ORDER_CUSTOMER_LINK_KEY
1	C4CA4238A0B923820DCC509A6F75849B	C4CA4238A0B923820DCC509A6F75849B	2025-07-29 13:22:03.000	6512BD43D9CAA6E02C990B0A82652DCA
2	ECCBC87E4B5CE2FE28308FD9F2A7BAF3	C81E728D9D4C2F636F067F89CC14862C	2025-07-29 13:45:01.000	6364D3F0F495B6AB9DCF8D3B5C6E0B01
3	C81E728D9D4C2F636F067F89CC14862C	ECCBC87E4B5CE2FE28308FD9F2A7BAF3	2025-07-29 13:45:01.000	37693CFC748049E45D87B8C7D8B9AACD

The task doesn't reinsert existing relationships.

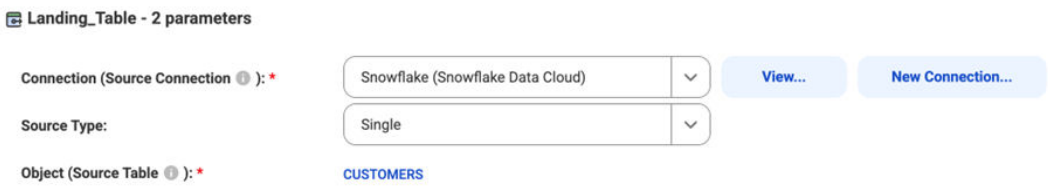
Step 3. Load customer metadata into satellite tables

We want to dynamically load customer metadata to a satellite table. We want to insert new records and update changed data. To do this, we use the Data Vault – Satellite mapping task template.

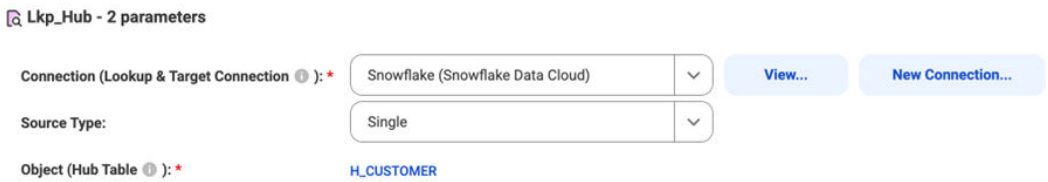
The following image shows the mapping this task is based on:



1. On the **Templates** page, click the **Data Vault – Satellite** tile and then click **Create a mapping task**.
2. Select the runtime environment to run the task and then click **Next**.
3. On the **Sources** tab, for the Landing_Table source, select the Snowflake connection and CUSTOMERS landing table object.



4. For the Lkp_Hub source, select the Snowflake connection and the hub table H_CUSTOMER.



5. The Lkp_Satellite source uses the same connection as the Lkp – Hub source. Select the S_CUSTOMER table.



6. On the **Targets** tab, configure the Trg_Satellite_Insert and Trg_Satellite_Update transformations.

   The following table shows how you configure the targets:

Transformation name	Connection	Object	Operation
Trg_Satellite_Insert	Snowflake	S_CUSTOMER	Insert
Trg_Satellite_Update	Snowflake	S_CUSTOMER	Update by CUSTOMER_HUB_KEY

7. On the **Input Parameters** tab, configure the following parameters:
   a. Hub Business Key. Defines the business key for the hub. You select CUSTOMER_HUB_KEY.
   b. Business Key. Defines the customer business key. You define the Src_CUSTOMER_ID field as the business key with the following expression:

      ```
      MD5 TO_CHAR(TRUNC(Src_CUSTOMER_ID))
      ```

   c. Lookup Satellite Condition. Performs a lookup on the satellite table. You define the lookup on the Satellite table based on the Hub key and the VALID_TO field using the following expression:

      ```
      CUSTOMER_HUB_KEY = Hub_Key_Calculated AND VALID_TO = VALID_TO_filter
      ```

   d. Change Compare. Compares new records to existing records. You want to check if the received data has any changes when compared to existing data. You enter the following expression:

      ```
      IIF( ISNULL( VALID_FROM ) , 'I' , IIF( Src_NAME || '~' || Src_ADDRESS || '~' || Src_CITY || '~' || Src_STATE || '~' || Src_POSTALCODE || '~' || Src_COUNTRY <> NAME || '~' || ADDRESS || '~' || CITY || '~' || STATE || '~' || POSTALCODE || '~' || COUNTRY , 'U' , 'X' ) )
      ```

   e. Insert Field Mapping. Defines the field mapping when new records are inserted.

      The following table shows how to map fields to the target:

Target field	Incoming field
CUSTOMER_HUB_KEY	Hub_Key_Calculated
NAME	Src_NAME
ADDRESS	Src_ADDRESS
CITY	Src_CITY
STATE	Src_STATE
POSTALCODE	Src_POSTALCODE

Target field	Incoming field
COUNTRY	Src_COUNTRY
VALID_FROM	VALID_FROM_INS
VALID_TO	VALID_TO_INS

f. Update Field Mapping. Defines the field mapping when records are updated.

The following table shows how to map fields to the target:

Target field	Incoming field
CUSTOMER_HUB_KEY	Hub_Key_Calculated
NAME	Src_NAME
ADDRESS	Src_ADDRESS
CITY	Src_CITY
STATE	Src_STATE
POSTALCODE	Src_POSTALCODE
COUNTRY	Src_COUNTRY
VALID_FROM	
VALID_TO	VALID_TO_UPD

8. Click **Save**.

When the task runs, it loads customer metadata into the satellite table as shown in the following image:



	CUSTOMER_HUB_KEY	NAME	ADDRESS	CITY	STATE	POSTALCODE	COUNTRY	VALID_FROM	VALID_TO
1	C4CA4238A0B923820DCC509A6F75849B	Alice Wonderland	123 Rabbit Hole Rdt	Fantasyland	CA	90210	USA	2025-07-29 16:23:20.000	9999-12-31 00:00:00.000
2	C81E728D9D4C2F636F067F89CC14862C	Bob The Builder	45 Construction Way	Builderville	TX	75001	USA	2025-07-29 16:23:20.000	9999-12-31 00:00:00.000
3	ECCBC87E4B5CE2FE28308FD9F2A7BAF3	Charlie Chaplin	789 Silent Film St	Hollywood	CA	90028	USA	2025-07-29 16:23:20.000	9999-12-31 00:00:00.000
4	A87FF679A2F3E71D9181A67B7542122C	Diana Prince	hemyscira Embassy, Suite 1	Washington	DC	20001	USA	2025-07-29 16:23:20.000	9999-12-31 00:00:00.000
5	E4DA3B7FBBCE2345D7772B0674A318D5	Eve Harrington	10 Broadway Ave	New York	NY	10001	USA	2025-07-29 16:23:20.000	9999-12-31 00:00:00.000
6	1679091C5A880FAF6FB5E6087EB1B2DC	Peter Parker	57 Broadway Ave	New York	NY	10002	USA	2025-07-29 16:23:20.000	9999-12-31 00:00:00.000

The task inserts a new record if the corresponding hub key is new or updates the satellite with a new version if the hub key exists and attributes have changed.

Author

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